Vaxcel®

CEFOTAXIME-1g

INJECTION

MAL20034201AZ

COMPOSITION:
Each vial contains
Cefotaxime Sodium equivalent to Cefotaxime 1g.

PRESENTATION:
A white to slightly yellow powder.

INDICATIONS:
Infection of respiratory tract, including throat, nose, ear; kidneys and urinary tract; skin and soft tissue; bones and joints; genital organs including gonorrhea and infection in the abdominal region. Bacterial endocarditis, septicemia, meningitis; for perioperative prophylaxis in patients who are at increased risk from infection, and for the prophylaxis of infections in patients with reduced resistance.

MECHANISM OF ACTION:
Cefotaxime is a third generation cephalosporin antibiotic with bactericidal effect. Its bactericidal activity results from inhibition of cell wall synthesis. It is used in the treatment of infections due to susceptible organisms especially serious and life threatening infections. It is highly stable to hydrolysis by most beta-lactamases and has greater activity against Gram-negative bacteria than first or second generation cephalosporins. In most pathogens, its minimum bactericidal concentration is slighty higher than its minimum inhibitory concentration. Desacethyl cefotaxime is an active metabolite of Cefotaxime and there may be additive or synergistic effects against Gram-negative bacteria, but not against *Pseudomonas spp.*

Spectrum of activity:
Cefotaxime generally effective against the following Gram-negative becteria: Staphylococci, aerobic and anaerobic Streptococci, Streptococcus pneumoniae, Neisseria sp, Haemophilus influenzae, Escherichia coli, Citrobacter sp, Salmonella sp, Klebsiella sp, Enterobacter aerogenes, Serratia sp, indole-positive and indole-negative Proteus sp, yersinia enterocolitica, Clostridium and Bacteroides spp.
Among the Gram-positive bacteria, Cefotaxime is active against staphylococci and streptococci. Pathogen with varying susceptibility are: Streptococcus faecalis, Enterobacter cloacea, Pseudomonas aeruginosa and Bacteroides fragilis. Cefotaxime is not effective against *Treponema pallidum* and *Clostridium difficile*.

Infections with *Pseudomonas aeroginosa* may require concomitant treatment with other antibiotics effective against Pseudomonas.

Combined treatment:
In severe, life threatening infections, the combination of Cefotaxime with aminoglycosides is indicated without awaiting the result of sensitivity tests. The 2 preparations must be administered separately, not mixed in 1 syringe.

Resistance:
Resistance may develop during treatment with Cefotaxime and is thought to be due to derepression of chromosomally-mediated beta-lactamases, although other mechanisms may be involved. It has been reported particularly in Enterobacter and multiresistant strains have emerged during treatment. Resistance has also develop in *Klebsiella*, *Serratia*, and *Pseudomonas spp.*

PHARMACOLOGY:
Cefotaxime is administered by injection as the sodium salt. It is rapidly absorbed after intra-muscular injection and mean peak plasma concentrations of about 12 and 20ug per ml have been reported 30 minutes after 0.5 and 1g of cefotaxime, respectivaly. Immediately after the intravenous injection of 0.5, 1 or 2gm of Cefotaxime mean peak plasma concentrations of 38,102 and 215μg per ml, respectively, have been achieved with concentrations ranging from about 1 to 3 μg per ml after 4 hours. The plasma half llife of Cefotaxime is about one hour and that of the active metabolite desacetyl cefotaxime is about 1.5 hours. Half life are increased in neonates and in patients with severe renal failure. About 40% of cefotaxime in the circulation is reported to be bound to plasma proteins. Cefotaxime and desacetyl cefotaxime are widely distributed in body tissue and fluids. Therapeutics concentrations are achieved in the cerebrospinal fluids particularly when the meninges are inflamed. It crosses the placenta and is excreted in low concentration in breast milk. Elimination is mainly by kidneys and about 40 to 60% of a dose has been recovered unchanged in the urine within 24 hours. Probenecid competes for renal tubular secretion with Cefotaxime resulting in higher and prolonged plasma concentrations of cefotaxime and its desacetyl metabolite. Cefotaxime and its metabolites are removed by haemodialysis. High concentrations of Cefotaxime and desacetyl cefotaxime are achieved relatively in bile and about 20% of a dose has been recovered in the faeces. There is not yet sufficient clinical experience with Salmonella typhi and paratyphi A and B infections.

DOSAGE AND ADMINISTRATION:
Cefotaxime is given as the sodium salt by deep intramuscular injection or intravenously by slow injection over 3 to 5 minutes or by infusion over 20 to 60 minutes. Doses are expressed in terms of the equivalent amount of cefotaxime.
Dosage and mode of administration and intervals between injections depend in the severity of the infection, sensitivity of the pathogens and condition of patients.

Type of infection	Daily dose	Dosage interval
Gonococcal urethritis/ cervicitis in males and females.	0.5g	0.5g IM (single dose)
Rectal gonorrhoea in females.	0.5g	0.5g IM (single dose)
Rectal gonorrhoea in males.	1g	1g IM (single dose)
Uncomplicated Infections	2g	1g every 12 hours IM or IV
Moderate to severe infections	3-6g	1-2gm every 8 hours IM or IV
Infections commonly needing antibiotics in higher dosage (eg septicemia)	6-8g	2gms every 6-8 hours IV
Life-threatening infections	Up to 12gm	2gm every 4 hours IV

Effective use for elective surgery depends on the time of administration. To achieve effective tissue levels, Vaxcel Cefotaxime should be given 1/2 or 1 1/2 hours before surgery. For patients undergoing gastrointestinal surgery, preoperative bowel preparation by mechanical cleansing as well as with a non-absorbable antibiotic (e.g., neomycin) is recommended. If there are signs of infection, specimens for culture should be obtained for identification of the causative organism so that appropriate therapy may be instituted.

Infants and children up to 12 years:
50-100mg/kg body weight daily, divided into several equal doses injected at intervals of 6-12 hours. For life-threatening infection, 150-200mg/kg can be given with the same intervals.

In premature infants, renal clearance is not yet fully developed; therefore daily doses of 50mg/kg body weight should not be exceeded.

Children > 12 years can be treated with adult dose.

Adult:
Unless otherwise prescribed, adults and children over 12 years are given 1g of Cefotaxime every 12 hours. In severe infections the daily dose may be raised to a maximum of 12g.

For the treatment of gohorrhoea:
A single dose of 500mg Cefotaxime should be administered IM. With less sensitive bacterial, dose should be increased. Before start the therapy, patient should be examined for syphilis.

For the perioperative prophylaxis of infections:
Administration of 1g Cefotaxime 30-60 minutes before the start of surgery is recommended. Depending on the risk of infection, the same dose may be repeated.

Dose in patients with impaired renal function:
In patients with a creatinine clearance ≤ 5ml min the maintenance dose should be reduced to half the normal dose. The initial dose depends on the sensitivity of the pathogens and the severity of the infections. Because high and prolonged serum antibiotic concentrations can occur from usual doses in patients with transient or persistent reduction of urinary output because of renal insufficiency, the total daily dosage should be reduced when Vaxcel Cefotaxime is administered to such patients. Continued dosage should be determined by degree of renal impairment, severity of infection, and susceptibility of the causative organism. Although there is no clinical evidence supporting the necessity of changing the dosage of Cefotaxime sodium in patients with even profound renal dysfunction, it is suggested that, until futher data are obtained the dose of cefotaxime sodium be halved in patients with estimated creatinine clearances of less than 20ml/min/1.73m². When only serum creatinine is available, the following formula² (based on sex, weight and age of the patient) may be used to convert this value into creatinine clearance. The serum creatinine should represent a steady state of renal function.

Males: $\dfrac{\text{Weight (kg) x (140-age)}}{\text{72 x serum creatinine}}$

Females: 0.85 x above value

Dose for Heamodialysis:
0.5-2g IV q24 after dialysis.

Duration of treatment:
The duration of treatment depends on the patient's response. It must be continued for at least 3 days after normalization of the body temperature.

Preparation of Vaxcel Cefotaxime Injection:

Strength	Diluent (mL)	Approximate Concentration (mg/mL)
1g vial (IM)	3	300
1g vial (IV)	10	95
1g infusion	50-100	20-10
2g infusion	50-100	40-20

Route of Administration:
Cefotaxime solutions should be administered as soon as they have been prepared. To be injected preferably by the IV route.

Intravenous injection:
For IV injection the contents of one vial of cefotaxime 1g are dissolved in at least 4ml of water for injection; the solution is then injected over a period of 3 to 5 times. 2gm are dissolved in at least 10ml water for injection.

Intravenous infusion:
If higher doses are required they may be administered by intravenous infusion. Short infusion 2g of Cefotaxime are dissolved in 40ml of water for injection or one of the usual infusion solution (e.g.normal saline, Ringer's solution, 5% dextrose solution, sodium lactate solution) and then administered over approximately 20 minutes. For continuous drip infusion, 2g Cefotaxime are dissolved in 100ml of one of the above mentioned infusion solutions and then administered over 50-60 minutes. *Sodium bicarbonate solution must not be mixed with Cefotaxime*. During post-marketing surveillance, potentially life-threatening arrythmia has been reported in very few patients who received rapid intravenous administration of cefotaxime through a central venous catheter.

Intramuscular administration:
1g of cefotaxime is dissolved in 4ml of water for injection; the solution is then injected deep into the gluteus muscle. Pain resulting from the IM injection can be prevented by dissolving Cefotaxime in the corresponding amount of 1% lignocaine solution, but in this case intravascular injection must be strictly avoided. Intravemous injection is recommended if the daily dose exceeds 2g or if Cefotaxime 1g is administered more than twice daily.

Reconstitution:
To avoid septic complications on injection, care should be taken during reconstitution to ensure aseptic handing. The solution should be used immediately after reconstitution. Aseptic handing is particularly important if the solution is not intended for immediate use. After reconstitution, Cefotaxime can be stored for up to 24 hours at temperatures below 25^{O}C without undergoing any significant phsical or chemical changes. A pale yellowish color of the solution does not indicate change in potency.

SIDE EFFECT:
The broad spectrum third generation cephalosporins have the potential for colonization and super infection with resistant organisms at various sites in the body, although the incidence has generally been low with Cefotaxime. Pseudomembranous colitis, associated with *Clostridium difficile* infection, may occasionally be seen with any of the third generation cephalosporins.

Effects on the Blood Picture:
Thromboctopenia, eosinophilia and leucopenia. As with other β - lactam antibiotics, granulocytopenia and, more rarely, agranulocytosis may develop during treatment with Cefotaxime, particularly if given over long periods. For courses of treatment lasting longer than 10 days, blood count should therefore be monitored. Rare cases of haemolytic anaemia have been reported.

Hypersensitivity reactions:
Allergic skin reactions (eg. urticaria, exanthema), itching, drug fever, interstitial nephritis (in rare cases) and anaphylaxis may occur. Anaphylactic shock is life-threatening and requires emergency treatment.

Effects on Liver Function:
Rise in serum bilirubin and / or liver enzyme level (SGOT, SGPT, y-GT, alkaline phosphatase, LDH).

Effects on renal Function:
A transient increase in serum creatinine and urea is possible.

Effects on the Gastrointestinal Tract:
Nausea and vomiting abdominal pain diarrhoea. The possibility of pseudomembranous colitis should be considered in patients in whom severe, persistent diarrhoea occurs during treatment or in the initial weeks thereafter. This intestinal inflammatory condition induced by antibiotic therapy may be life-threatening. Administration of cefotaxime is to be halted immediately in such circumstances. The doctor must initiate appropriate treatment without delay. Drugs that inhibit intestinal motility (peristalsis) must not to be taken in such cases.

Local Reactions:
Inflammatory irritation and pain at the site of injection.

Other reactions:
As with other antibiotics, the use of Cefotaxime, especially if prolonged, may result in overgrowth of non-susceptible organisms. Repeated evaluation of the patient's condition is essential. If superinfection occurs during theraphy, appropriate measures should be taken. The occurence of one or more of the following symptoms has been reported after several week's treatment of borreliosis: Skin rash, itching, fever, leucopenia, increase in liver enzymes, diffficulty in breathing, joint discomfort. To some extent, these manifestations are consistant with the symptoms of the underlying disease of which the patient is being treated. An increase in serum liver enzymes (eg. ALAT, ASAT, LDH, gamma-GT and/or alkaline phospatase and/or bilirubin may be observed. This increase which may also be explained by the infection itself, only rarely exceeds twice the upper limit of the normal range and may be a manifestation of a usually cholestatic liver injury, which, in most cases, is asyptomatic). Decrease in renal function (increase in serum creatinine) particularly when co-prescribed with aminoglycosides. As with some other cephalosporins in rare cases, intersitial nephritis may occur. Cardiovascular: In isolated cases, arrythmia has been observed following rapid bolus infusion through a central venous catheter. As with other cephalosporins, isolated cases of bullous eruptions (erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis) have been observed. Administration of high doses of beta-lactam antibiotics may result in encepalopathy. (with impairment of consciousness, abnormal movements and convulsions). As reported with other antibiotics, for the treatment of borreliosis infections, Jarisch-Herxheimer's reaction characterised by the occurence or worsening of general symptoms such as a fever, chills, headache and joint pains may develop during the first days of treatment. Please consult a physician if notice any of the adverse effects listed in this package insert or any other undesired effects or unexpected changes. Since some adverse effect may be life threatening (eg. pseudomembranous colitis, anaphylaxis, change in blood picture), inform your physician at once.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE:
Emergency measures to be taken in the event of anaphylactic shock:
The following emergency measures are generally recommended:
At the first signs (sweating, nausea, cyanosis) interrupt of the injections immediately but leave the venous cannula in place or perform venous cannulation. In addition to the usual emergency measure, ensure that the patient is kept flat with the legs raised and airways patent.
Emergency Drug Therapy:
Immediate epinephrine (adrenaline) IV:
Dilute 1 ml of epinephrine solution 1:1000 to 10ml. In the first instance slowly inject 1ml of this dilution (equivalent to 0.1mg epinephrine) while monitoring pulse and blood pressure (watch for disturbances of cardiac rhythm). The administration of epinephrine may be repeated. The glucocorticoids IV, eg. 250-1000mg mg methylprednisolone. The glucocoticoids administration may be repeated. Subsequently volume substitution IV, eg. plasma expenders, human albumin, balanced electrolyte solution.

Other Therapeutic Measures:
Artificial respiration, oxygen inhalation, calcium, antihistaminics.

PRECAUTIONS:
In patients hypersensitive to penicilins, the possibility of cross-sensitivity exists. Renal function must be monitored in case of combination with aminoglycosides. For courses of treatment lasting longer than 10 days, the blood count should be monitored and treatment with Cefotaxime should be stopped in the event of neutropenia. In cases of severe and persistent diarrhoea, antibiotic-induced pseudomembranous colitis should be suspected which may be life-threatening. In such cases, Cefotaxime should be discontinued immediately and suitable treatment initiated, eg. vancomycin oral, Peristalsis-inhibiting drugs are contraindicated. The prescription of cephalosporins necessitates preliminary enquiry with regard to hypersensitivity to beta-lactam antibiotics. The possibility of cross-sensitivity exists. The dosage should be modifed according to the creatinine clearance calculated. If necessary on the basic of serum creatinine. Administration of antibiotics, especially if prolonged, may lead to the proliferation of resistance microorganisms. The patient's condition must therefore be checked at regular intervals. If a secondary infection occurs appropriate measures must be taken. The sodium content should be taken into consideration. Some adverse effect may impair the ability to concentrate and react and therefore constitute a risk in situations where these abilities are of particular importance (operating a machinery or vehicle).

PREGNANCY AND LACTATION:
Cefotaxime should not be used during pregnancy, especially the first trimester, unless strictly indicated. As Cefotaxime is excreted in breast milk, either breast-feeding or treatment of the mother with Cefotaxime should be discontinued.

Warning:
Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Vaxcel Cefotaxime-1g Injection, careful inquiry should be made concerning previous hypersensitivity reactions to Cefotaxime sodium, penicillins, cephalosporins, carbapenems or other beta-lactam agents. This product should be given with caution to patients with type I hypersensitivity. Antibiotics should be administered with caution to any patient who has demonstrated some form of allergy, particularly to drugs. If an allergic reaction occurs, Vaxcel Cefotaxime-1g Injection must be discontinued immediately and appropriate alternative therapy instituted. Serious hypersensitivity reaction may require epinephrine and other emergency measure.

DRUG INTERACTION:
By delaying renal excretion, the concurrent administration of probenecid increase the concentration of Cefotaxime in serum and prolongs its duration of action. Increased nephrotoxicity has been reported following concomitant administrationof cephalosporins and aminoglycoside antibiotics. Administration of cephalosporins, including Cefotaxime, may cause a transient reduction in plasma concentrations of oestrogens and gestagens. The effectiveness or oral contraceptives is therefore uncertain.

Interference with laboratory test:
The direct Coomb's test may give false-positive result in patients on Cefotaxime treatment. False-positive results may be obtained for urinary glucose if it is determined by reduction methods (non-anzymatic method); this can be avoided by using enzymatic methods in patients under Cefotaxime treatment.

CONTRAINDICATIONS:
Hypersensitivity to Cephalosporins.

INCOMPATIBILITIES:
Cefotaxime sodium should not be mixed with alkaline solutions such as sodium bicarbonate injection or solutions containing aminophylline. Cefotaxime should not be admixed with aminoglycosides. If they are used concurrently they should be administered in separate sites. Cefotaxime should not be mixed with other diluents except those mentioned in route of adminstration.

STORAGE:
Keep container well closed. Stored below 30^{O}C. Protect from light. The freshly reconstituted solution is recommended.

PACK QUANTITIES:
Available in 1 vial per box and 10 vials per box.

Further information can be obtained from your pharmacist, physician or the manufacturer.

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